Yixuan Xie

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Education

University of Illinois at Urbana-Champaign *08/2026 -- Present*
M.S. in Computer Science -- Advisor: Prof. Jiawei Han *GPA: 4.0/4.0*

University of Illinois Urbana-Champaign *08/2022 -- 05/2026*
B.S. in Computer Science & Statistics -- Advisor: Prof. Jiawei Han *GPA: 3.9/4.0*

Publications

† indicates equal contribution.

MemComposer: Learning to Assemble Memories for LLM Agents via Reinforcement Learning
Yixuan Xie, Xueqiang Xu, Xinyi Fan, Xinxi Lyu, Tao Feng, Xing Fan, Chenlei Guo, Jie Hao, Shaowen Wang, Jiashuo Sun, Jiawei Han.
In submission to ARR Oct, 2026

CaseMem: Closing the Loop from Agent Outcomes to Memory Reuse
Xinyi Fan, **Yixuan Xie**, Xing Fan, Chenlei Guo, Jie Hao, Jiawei Han.
In submission to ICLR, 2027

TaSR-RAG: Taxonomy-guided Structured Reasoning for Retrieval-Augmented Generation
Jiashuo Sun†, **Yixuan Xie**†, Jimeng Shi, Shaowen Wang, Jiawei Han.
In submission to ARR Oct, 2026 [arXiv] [Code]

Retrieval is Cheap, Show Me the Code: Executable Multi-Hop Reasoning for RAG
Jiashuo Sun†, Jimeng Shi†, **Yixuan Xie**, Saizhuo Wang, Jash Rajesh Parekh, Pengcheng Jiang, Zhiyi Shi, Jiajun Fan, Qinglong Zheng, Peiran Li, Shaowen Wang, Ge Liu, Jiawei Han.
In submission to ICLR, 2027 [arXiv] [ProjectPage] [Code]

MultiCube-RAG: Training-Free Multi-Hop RAG by Reasoning over Structured Knowledge Cubes
Jimeng Shi, Wei Hu, Runchu Tian, Bowen Jin, Wonbin Kweon, SeongKu Kang, Yunfan Kang, Dingqi Ye, Sizhe Zhou, Jiashuo Sun, **Yixuan Xie**, Shaowen Wang, Jiawei Han.
In submission to AAAI, 2027 [arXiv]

Experience

Amazon-Illinois AICE | *Research Assistant* *05/2026 - Present*
Agent Memory Systems *Champaign, IL*

- Proposed **MemComposer**, an RL-trained memory composer that selects and orders memories for LLM agents, outperforming MemRL and Memento on τ^2-bench (+5.9/+4.6 pp avg. task success) with up to 17% fewer memory tokens.
- Introduced **CaseMem**, an outcome-adaptive framework that updates reuse preference based on downstream task's result.

UIUC Data Mining Group | *Undergrad Researcher* *08/2025 - 05/2026*
Retrieval-Augmented Generation & RL -- Advised by Prof. Jiawei Han *Champaign, IL*

- Built **PyRAG**, reformulated multi-hop RAG as executable Python programs with compiler-grounded self-repair.
- Proposed **TaSR-RAG**, a taxonomy-guided structured reasoning framework for multi-hop QA; +14% over baselines.

Matsuo-Iwasawa Lab | *Research Assistant* *07/2025 - 12/2025*
LLM Post-Training -- Mentored by Dr. Qian Niu *Remote*

- Fine-tuned Qwen3 series models with SFT and GRPO, enhanced reasoning on medical QA tasks.

IBM-Illinois Discovery Accelerator Institute | *Research Assistant* *05/2025 - 08/2025*
Open-Schema Information Extraction -- Advised by Prof. Jiawei Han *Champaign, IL*

- Benchmarked training-free IE methods and LLM in-context learning across DocEE, MAVE, and NERRE.
- Conducted cross-method error analysis to compare robustness across open-schema IE settings.

DataLynn, Inc. | *Machine Learning Engineer Intern* *05/2024 - 08/2024*
Retrieval-Augmented Generation *Long Island City, NY*

- Worked on RAG pipelines for AI agents in customer support and interview prep.

Selected Projects

Token-Perplexity Signals for Stable RL Post-Training | *Team Lead* *09/2025 - 12/2025*
Machine Learning System Course Project -- Mentored by Prof. Fan Lai [Tech Report]

- Investigated gradient-based and token-level diagnostic signals for RLHF dynamics, implementing pipeline instrumentation to capture objective-specific gradients and prefix perplexity during GRPO training.
- Systematically analyzed gradient similarity and perplexity distributions across model scales, revealing limited sensitivity of gradient conflict and substantial overlap in perplexity between correct and incorrect rollouts.

Taxonomy-Enhanced Scientific Retrieval | *Group Lead* *01/2025 - 05/2025*
CS497 Team Project -- Advised by Prof. Jiawei Han [Tech Report]

- Investigated taxonomy-enhanced retrieval methods for scientific papers using LitSearch dataset (64K docs).
- Developed and evaluated multiple retrieval strategies—including taxonomy alignment, fusion embedding, and a fine-tuned encoder—demonstrating an improvement in Recall@5 from 39.0% to 47.5% (+8.5%).

Technical Skills

Languages: Python | C/C++
Core Frameworks & Tools: PyTorch | vLLM | veRL | LlamaFactory | Slurm
Systems & Training: ZeRO/DeepSpeed | Retrieval-Augmented Generation | RLHF/RLVR
Others: Docker | MySQL | MongoDB | Kubernetes | FastAPI | Milvus | Flask | Neo4j

Activities & Honors

- **Hackathons** – (2024 & 2025)
- **Datathons** – (2024 & 2025)
- **Course Assistant** – (08/2023 - 12/2023)
- **Dean's List** – (Spring 2025, Fall 2024, Spring 2023)

Last updated: Oct 13, 2026